

Auto MPG Prediction Report

1. Dataset Overview

This project uses the Auto MPG Dataset to predict the fuel efficiency of cars. The target variable is MPG (Miles Per Gallon), which represents how many miles a car can travel using one gallon of fuel.

2. Data Cleaning and Exploration

The dataset was loaded using pandas. Missing values, especially in the horsepower column, were handled by removing rows containing null values. Basic visualizations were created, including a histogram to show the distribution of MPG and a boxplot to detect possible outliers.

3. Data Splitting

The data was split into 70% training data and 30% testing data. The training data was used to train the models, while the testing data was used to evaluate their performance on unseen data.

4. Models Used

Several regression models were tested: Linear Regression, Polynomial Regression with degrees 2, 3, and 4, Ridge Regression, and Lasso Regression.

5. Model Evaluation

The models were evaluated using MSE, MAE, RMSE, MAPE, and R^2 Score. The main metric used to select the best model was RMSE, because it measures the prediction error in the same unit as the target variable.

6. Final Model Selection

After comparing the Polynomial Regression models, Degree 3 was selected because it provided the best balance between accuracy and model complexity. Then, Ridge Regression was applied to the Degree 3 polynomial model to reduce overfitting and improve generalization.

7. Ridge Regression Tuning

StandardScaler was used before Ridge Regression to scale the features. GridSearchCV was then used to choose the best alpha value from different options. This helped improve the final model performance.

8. Residual Analysis

A residual plot was created for the final Ridge model. The residuals were plotted against the fitted values to check whether the errors were randomly scattered around zero. A random pattern around zero indicates that the model fits the data well.

Summary of Final Choice

Item	Selected / Used
Best Polynomial Degree	Degree 3
Final Model	Polynomial Degree 3 + Ridge Regression
Main Selection Metric	Lowest RMSE
Purpose of Ridge	Reduce overfitting and improve generalization

Conclusion

Based on the evaluation results, **Polynomial Regression with Degree 3** was selected as the best polynomial model. **Ridge Regression** was then applied to improve the model generalization and reduce overfitting. Therefore, the final model used for prediction was **Polynomial Degree 3 + Ridge Regression**.